

BALAKUMARAN K

Full-Stack Developer

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PROFESSIONAL SUMMARY

Entry-level MERN Stack Developer with hands-on experience in building responsive **frontend interfaces** and secure **backend systems** using MongoDB, Express.js, React.js, and Node.js. Proficient in **REST API development**, JWT authentication, and **error handling**, with expertise in search, filtering, sorting, pagination, and **Swagger/OpenAPI** documentation. Familiar with React Query, Zustand, Tailwind CSS, Razorpay, **Socket.IO**, Cloudinary, Git, and deploying full-stack applications focused on performance, usability, and maintainable code.

SKILLS

Programming Languages : JavaScript, TypeScript, Java
Frameworks & Libraries : React, Node.js, Express, Tailwind CSS
Tools & Platforms : Git, GitHub, Postman, Linux
Databases : MongoDB, PostgreSQL, Mongoose, Indexing
Auth & Security : JWT, OAuth, RBAC, Session Management, bcrypt, Helmet, CORS, Rate Limiting

PROJECTS

MarketFlow - MERN E-Commerce Platform | 🌐 Github Jun 2026 - Jul 2026

- **Developed** a full-stack e-commerce platform utilizing **React.js, Node.js, Express.js, MongoDB, and Mongoose**, which includes authentication, product management, cart functionality, wishlist, orders, reviews, coupons, payments, and admin modules.
- **Created** responsive customer and admin interfaces with **React.js, Tailwind CSS, Zustand, TanStack Query**, and **Axios**, implementing protected routes, reusable components, and features such as search, filtering, sorting, and pagination.
- **Secured** REST APIs through **JWT authentication, bcrypt**, role-based and ownership authorization, validation, **Helmet**, CORS restrictions, input sanitization, rate limiting, and centralized error handling.
- **Integrated** Razorpay for payment processing and **Cloudinary** for image management; **documented** over 25 APIs using **Swagger/OpenAPI**.

Technologies / Tools Used : React, Node.js, Express, MongoDB, Tailwind CSS, Razorpay, Zustand, TanStack Query

Talkative - Real-Time MERN Chat Application | 🌐 Github 🌐 Live

- Developed a **full-stack real-time chat application** using the **MERN** stack and **Socket.IO**, enabling **one-to-one conversations**, group chats, instant message delivery, typing indicators, online presence, and multi-device connections.
- Implemented **secure authentication** and authorization with **JWT, bcrypt**, protected REST APIs, **Socket.IO** authentication middleware, chat membership validation, and administrator-controlled group operations.
- Built a **reliable messaging system** with MongoDB persistence, featuring optimistic UI updates, acknowledgement-based delivery, client-generated message IDs for idempotency, cursor-based message pagination, unread-message tracking, and real-time notifications.
- Integrated **image and file sharing** using Multer and Cloudinary, and created a responsive React interface with mobile chat navigation, reusable components, centralized state management, error handling, loading states, and user-friendly feedback.

Technologies / Tools Used : React.js, Node.js, Express.js, MongoDB, Mongoose, Socket.IO, Multer, Cloudinary

EDUCATION

University College of Engineering | Ariyalur & Thirukkuvalai
Bachelor of Engineering in Computer Science and Engineering

2022 - 2026
CGPA : 7.69/10